

# Menjelajahi Peluang: Memahami Tingkat Kemungkinan m

Created: 6/2/2026

---

## Student

Student: Kheyila

Topic: Menjelajahi Peluang: Memahami Tingkat Kemungkinan melalui Eksperimen dan Simulasi

Status: completed

## Lesson Plan

### Learning Objectives

- Classify events using probability terms including certain, likely, even chance, unlikely, and impossible.
- Model the likelihood of events by placing them accurately on a probability scale from 0 to 1.
- Execute a simple probability experiment and record data systematically using a tally chart.
- Compare experimental results with theoretical predictions to explain the impact of trial frequency on accuracy.

### Lesson Information

## Lesson Sections

### 1. Spiral Review Warm-up (10 minutes)

Teacher Script:

Hi Kheyila! Great to see you today. Before we dive into our new adventure, let's flex those brain muscles with a quick review. Check out my screen! First, look at these Base-Ten blocks. Question 1: What number is shown by 4 thousands cubes, 2 hundreds flats, and 5 units? Type it in the chat! Question 2: If I have the number 7,038, which digit is in the tens place? Now, let's play a quick number game. Question 3: What is the largest 4-digit number you can make using 0, 9, 2, and 5? Lastly, Question 4: If I add 1,000 to 5,421, what is my new total? You're doing great!

Activities:

- *Interactive whiteboard visual: Representing 4,205 with blocks.*
- *Chatbox speed round for place value and number patterns.*

## 2. Introduction & Real-World Connection (5 minutes)

Teacher Script:

Perfect! Today, we are going to become 'Probability Explorers.' Have you ever wondered why some games are harder to win than others? Or why a weather forecaster says there is a 50% chance of rain? Probability is the math of 'guessing' the future! We use it in video games, sports, and even when deciding to bring an umbrella. Today, we'll learn the language of chance.

Activities:

- *Discussion on real-world examples: Weather, board games, and sports.*
- *Visual slide showing a weather app prediction.*

## 3. Review of Prior Knowledge (5 minutes)

Teacher Script:

Think back to when you've played games with dice or coins. If I toss a coin, can it land on its edge? Probably not! It's usually heads or tails. Does anyone know what we call something that will DEFINITELY happen? Or something that can NEVER happen? Let's see if we can label some everyday events on our chat.

Activities:

- *Brief brainstorm session on 'Possible' vs 'Impossible' events.*

## 4. Problem Introduction (10 minutes)

Teacher Script:

Look at this 'Likelihood Scale' on my screen. It goes from 0 (Impossible) to 1 (Certain). In the middle at 0.5, we have 'Even Chance.' Let's play a drag-and-drop game. I have five Situations: 1. The sun will rise tomorrow. 2. You will grow wings and fly. 3. Rolling a 7 on a standard 6-sided die. 4. Picking a red ball from a bag of 5 reds and 1 blue. 5. Tossing a coin and getting heads. Kheyla, drag these to where you think they belong on the scale!

Activities:

- *Drag-and-drop activity: Categorizing events on a 0-1 scale.*
- *Vocabulary matching: Linking terms like 'Likely' and 'Unlikely' to numerical zones.*

## 5. Guided Practice (15 minutes)

Teacher Script:

Now, let's test our predictions! We are going to conduct a 'Virtual Coin Flip' experiment. We predict that since it's an 'Even Chance,' we should get heads about half the time. I want you to flip the virtual coin 10 times. Fill out this tally chart on the screen for every flip. ... What happened? Did you get exactly 5 heads? Usually, with only 10 flips, it might be 6-4 or 7-3. Now, let's use the simulator to flip it 1,000 times instantly. Look at the percentage! It's getting closer to exactly 50%, right? That's the secret: more trials mean more accuracy!

Activities:

- *Conducting 10 manual trials of a coin flip.*
- *Using a digital simulator for 1,000 trials.*
- *Recording results in a Tally Chart.*

## 6. Independent Practice (15 minutes)

Teacher Script:

Your turn, Explorer! I've given you a virtual bag. Inside, there are 8 Green Marbles and 2 Yellow Marbles. First, tell me: Is it Likely, Unlikely, or Impossible to pick a Yellow marble? Now, simulate 20 picks. Record your data in a table. Then, answer this: Did your result match your prediction? Finally, if you added 10 more Yellow marbles, how would the probability change?

Activities:

- *Probability Simulation: Marble drawing.*
- *Data analysis and comparison between prediction and result.*
- *Problem-solving: Changing the variables to see the effect on probability.*

## 7. Consolidation & Reflection (5 minutes)

Teacher Script:

What a hunt today, Kheyla! You've learned that math can predict the likelihood of the future. We learned about the scale from 0 to 1 and how experiments help us prove our theories. Your homework is ready to keep these skills sharp. Don't forget to join <https://learn.algonova.id/login> to complete your tasks. Do you have any questions about our experiment today?

Activities:

- *Reflection: Name one thing that is 'Certain' to happen tomorrow.*
- *Overview of homework tasks.*

## Homework

### Medium Questions:

1. A bag has 4 red, 4 blue, and 4 green candies. What is the likelihood of picking a blue candy? (Use the correct term).
2. Place the following on a probability scale of 0 to 1: Picking a King from a deck of cards (Unlikely).
3. If you roll a standard 6-sided die, which term describes the chance of rolling a number less than 7?
4. A spinner has 5 sections: 1 is Gold, 4 are Silver. Is it 'Likely' or 'Unlikely' to land on Gold?

### Hard Questions:

1. You flip a coin 50 times and get 35 Heads. Is this result close to the theoretical 'Even Chance'? Explain why this might happen.

2. Design an experiment: If you wanted an event to be 'Impossible,' what would you put in a mystery box labeled 'Fruit'?
3. A bag contains 10 marbles. If the probability of picking a blue marble is 'Likely,' what is the minimum number of blue marbles that must be in the bag?
4. Explain the difference between 'Experimental Probability' (what happened in our coin flip) and 'Theoretical Likelihood' (what we expected).
5. Create a scenario that would be placed at exactly 0.75 on the probability scale.
6. If you spin a spinner with 8 equal sections (numbered 1-8), what is the likelihood of landing on an even number? Explain your choice.

**Answer Key:**

- `Medium 1: Even Chance ( $4/12 = 1/3$  is actually 'Unlikely' or 'Less than Even'. Wait, 4 out of 12 is  $1/3$ , so 'Unlikely' is the best fit.)`
- `Medium 3: Certain (All numbers 1-6 are less than 7).`
- `Hard 1: No, it is not close. This might happen due to the small number of trials or 'luck,' but with more trials, it should move toward 25.`
- `Hard 3: 6 marbles (Must be more than half/5 to be 'Likely').`
- `REVIEW: Place Value - 4 thousands, 2 hundreds, 0 tens, 5 units = 4,205.`
- `REVIEW: Number Games - Largest number from 0, 9, 2, 5 is 9,520.`

**Parent Reminder**

**Today's Math Adventure: Probability!**

Hi! Today Kheyly explored the 'Science of Chance.' She learned how to predict if events are certain or impossible and ran her own coin-flip experiments. We also reviewed Place Value to keep her foundations strong. Please encourage her to check her dashboard at <https://learn.algonova.id/login> for her fun exploration homework!